

SCREW ELEMENTS 工艺单元螺纹原件

➤ Extruder (Mechanics)

- Introduction
- Overview
 - ZSE
 - Drive
 - Processing unit

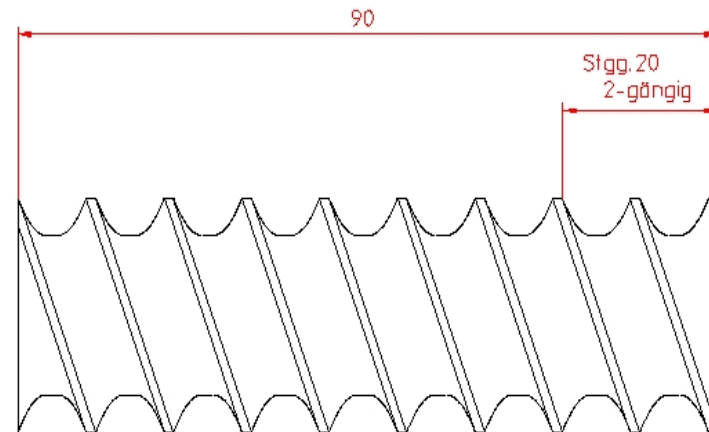
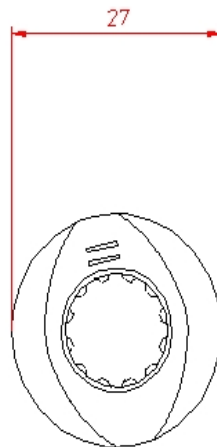
Gleichlauf-Fördererelement Auskämmend

co-rotating

conveying

intermeshing

GFA_2_20_90



PROCESSING UNIT

SCREW ELEMENTS 工艺单元螺纹原件

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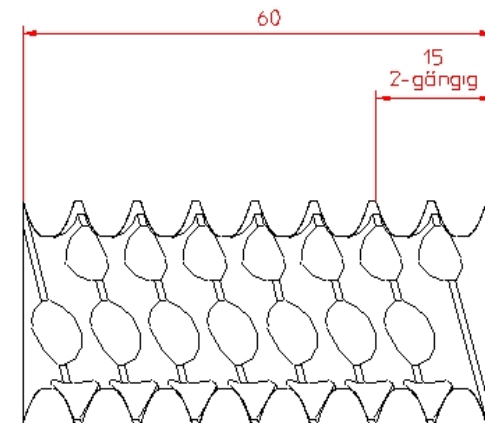
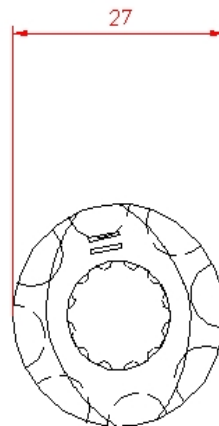
Gleichlauf-Fördererelement Mischend

co-rotating

conveying

mixing

GFM_2_15_60



PROCESSING UNIT

SCREW ELEMENTS 工艺单元螺纹原件

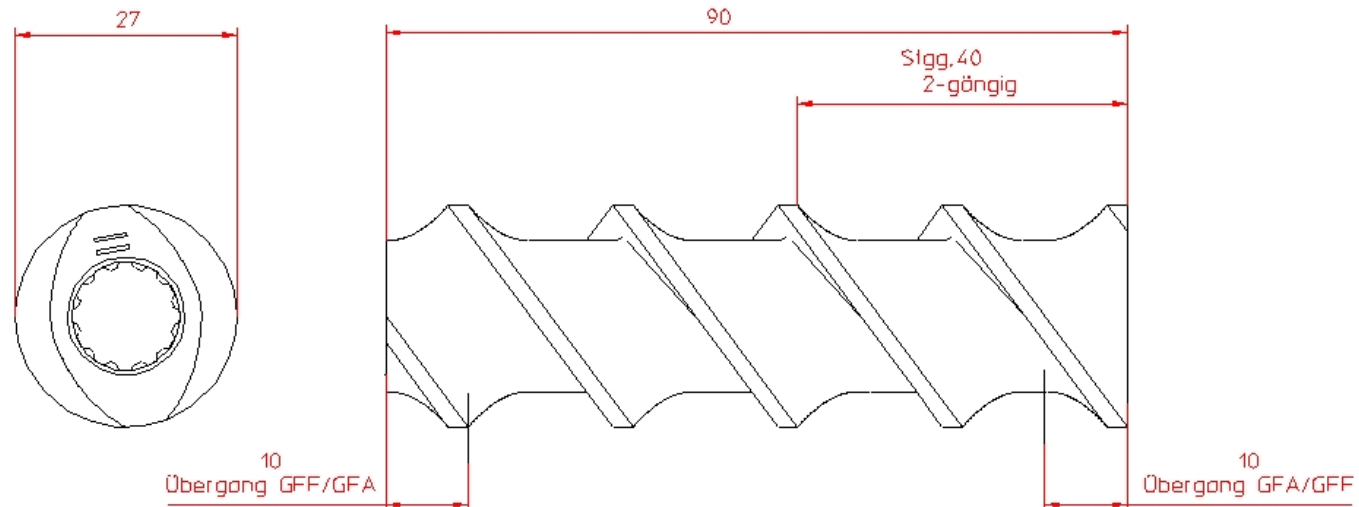
Gleichlauf-Fördererelement Freigeschnitten

co-rotating conveying non-self-wiping (high free volume)

GFF_2_40_90

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PROCESSING UNIT

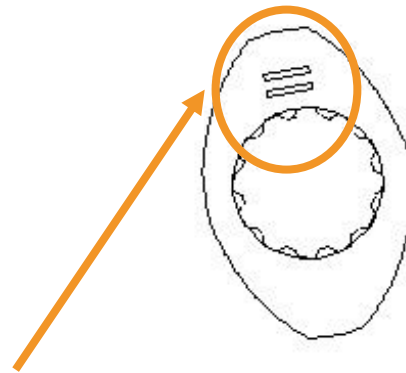
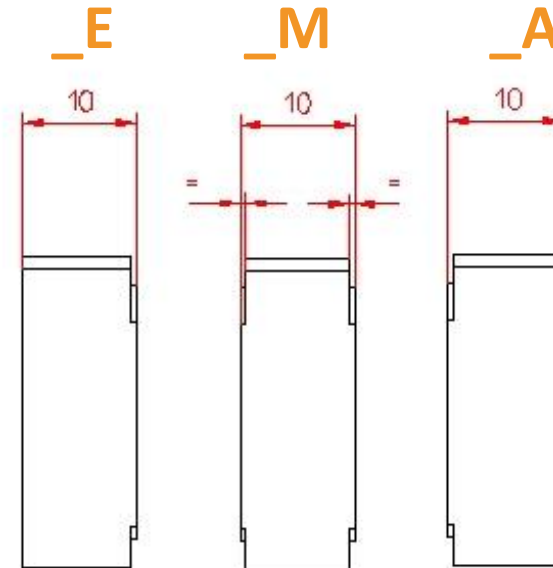
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KnetScheibe / kneading disc

KS1_2_10



material type 材料类型（字面向变速箱）
(stamping faces the gear)

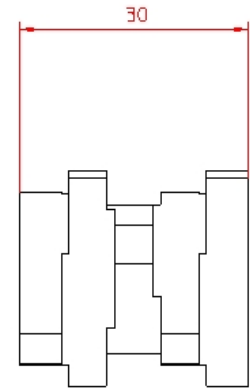
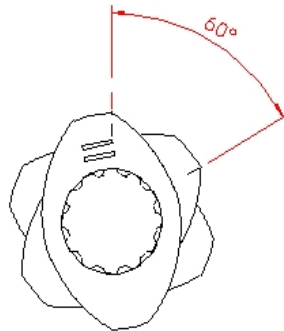
➤ Extruder (Mechanics)

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KnetBlock / kneading block

KB5_2_30_60° similar to kneading disc arrangement

KS5_2_30_60° 与KS原件很像



SCREW ELEMENTS 工艺单元螺纹原件

Characterization and mode of operation of screw elements: 螺杆原件的特性和操作方式

➤ Extruder

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- **Pitch** (geometrical quantity) influences the chamber volume and residence time
导程（几何量）影响自由容积和滞留时间
- **Chamber volume** describes the available volume in the screw flight
自由容积表示在螺纹线间的可用空间
- **Self-cleaning** becomes possible by the „Erdmenger profile“ of the screws and is ideal for conveying elements
“艾德蒙哥剖面”使得自清洁可行并且是理想的输送原件
- **Residence time** of the material in the screw is determined by the pitch (the higher the pitch, the shorter the residence time) <> but the residence time in the extruder is determined by the screw speed and the length of the processing unit!
停留时间由导程决定，但也与螺杆速度及机筒长度有关
- **Conveying effect** describes the element's ability to push the material forward. The more flights there are and the smaller the pitch is, the stronger the conveying effect.
传输效率表征原件推动材料向前的能力。螺纹线越多，导程越小，输送能力越强

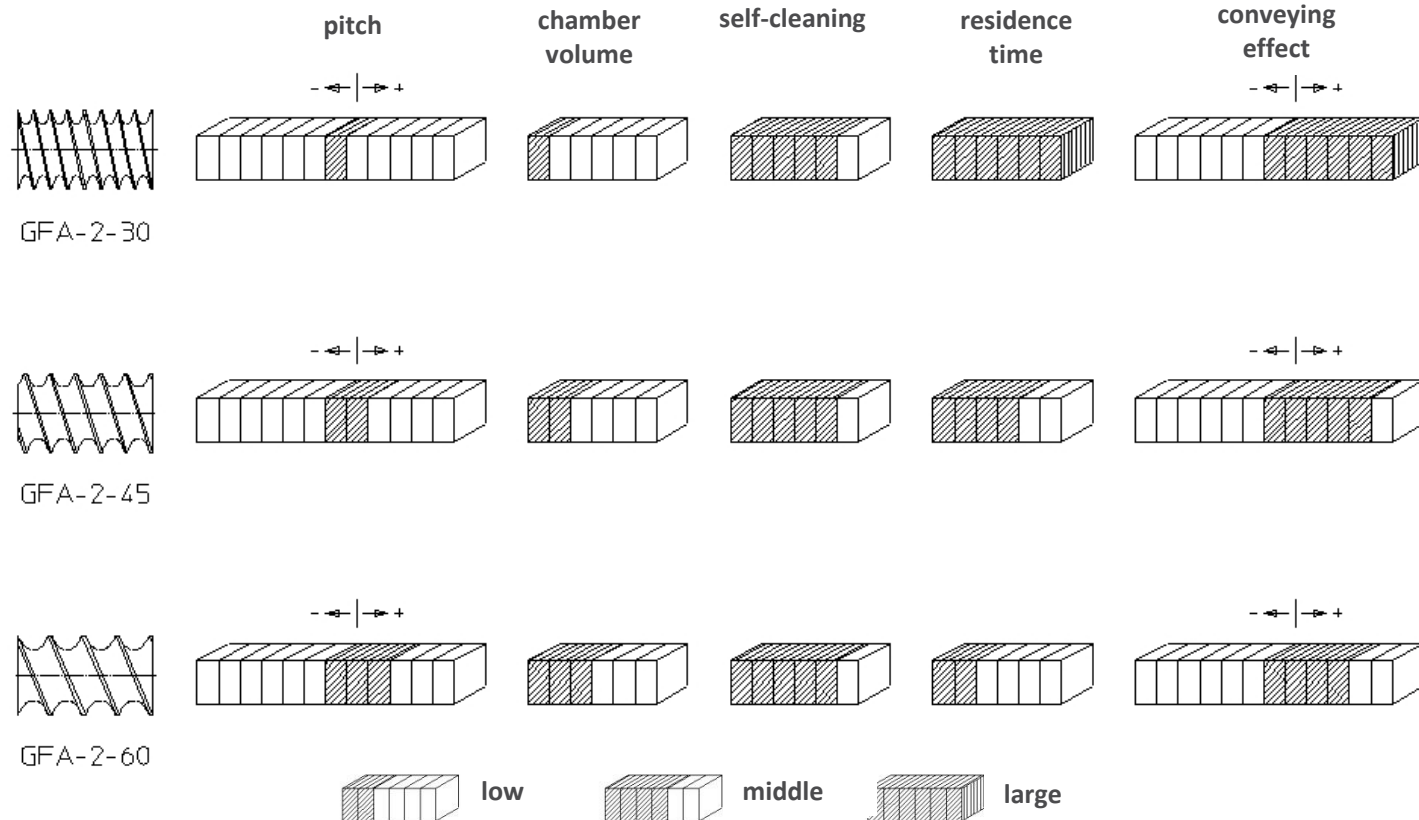
PROCESSING UNIT

SCREW ELEMENTS 工艺单元螺纹原件

Function of screws 原件的功能

Extruder (Mechanics)

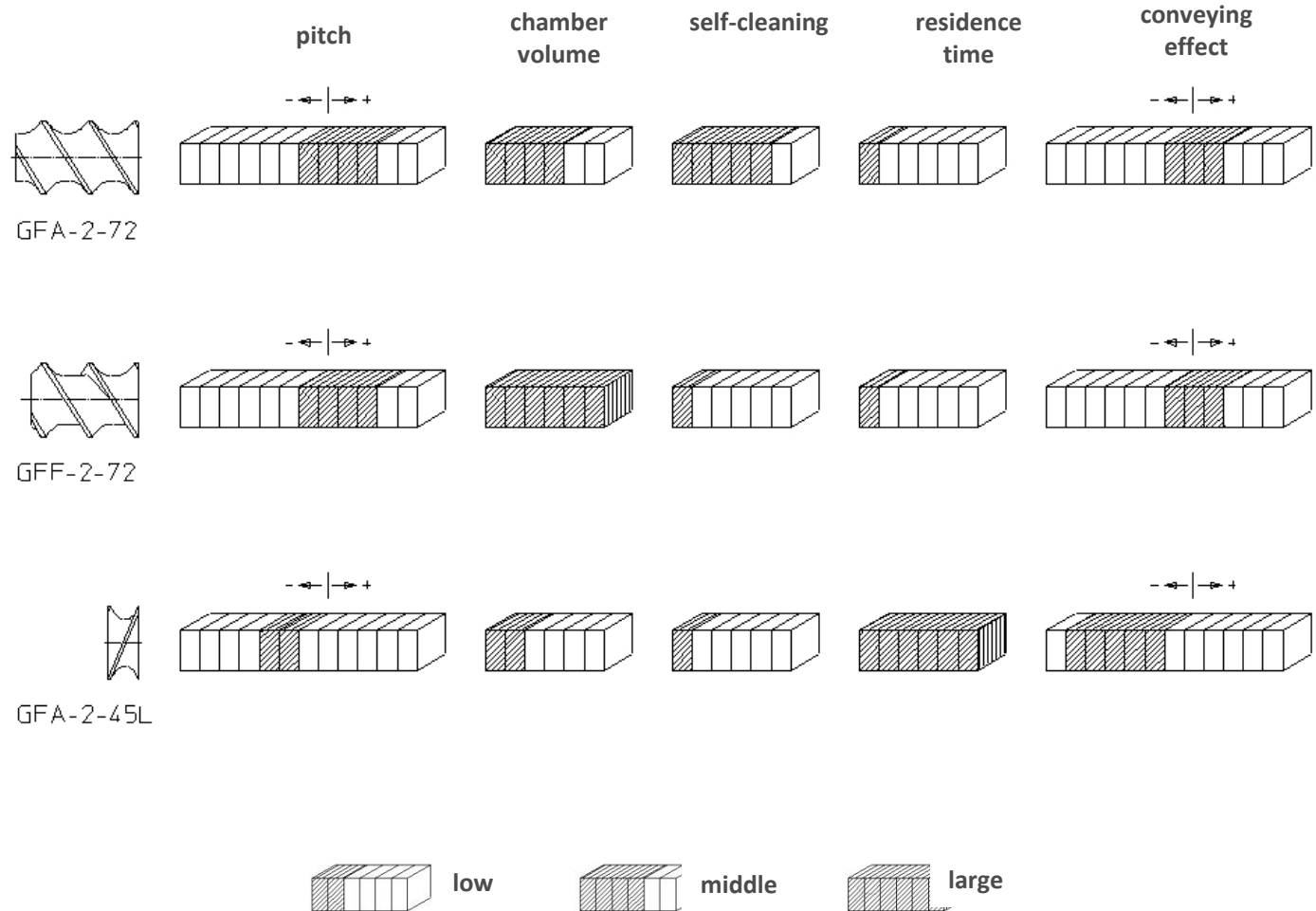
- Introduction
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Function of screws 原件的功能

➤ Extruder (Mechanics)

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➤ Extruder (Mechanics)

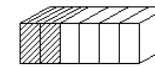
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**kneading bloc combination
30° conveying**

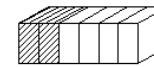


KB6-60-30°-RE

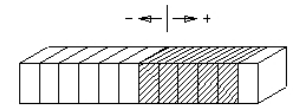
mixing



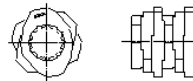
shearing



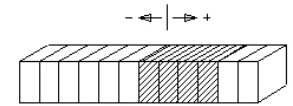
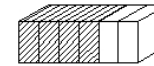
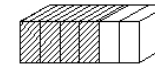
conveying



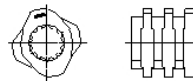
**kneading bloc combination
60° conveying**



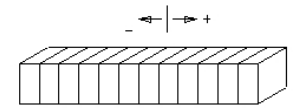
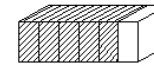
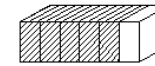
KB6-60-60°-RE



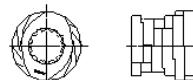
**kneading bloc combination
90°**



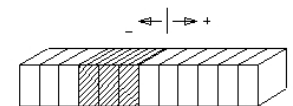
KB6-60-90°



**kneading bloc combination
30° reverse conveying**



KB6-60-30°-LI



low



middle



large

➤ Extruder (Mechanics)

➤ Introduction

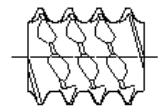
➤ Overview

➤ ZSE

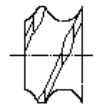
➤ Drive

➤ Processing
unit

Function of screws 原件的功能



GFM-2-30-60



GFA-2-45L-NUT

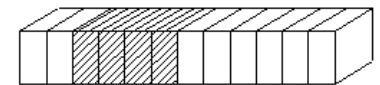
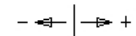
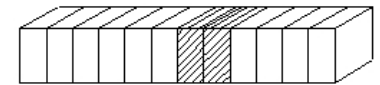
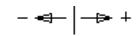
mixing



shearing



conveying



low



middle



large